What is a digital image?

A digital image may be defined as a two-dimensional function f(x,y), where 'x' and 'y' are spatial coordinates and the amplitude of 'f' at any pair of coordinates is called the intensity of the image at that point. When 'x,' 'y' and amplitude values of 'f' are all finite discrete quantities, the image is referred to as a digital image.

Digitising the coordinate values is referred to as 'sampling,' while digitising the amplitude values is called 'quantisation.' The result of sampling and quantisation is a matrix of real numbers.

A digitised image is represented as:

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \vdots & \vdots & \vdots & \vdots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

Each element in the array is referred to as a pixel or an image element.

Basic image processing commands in MATLAB

In MATLAB a digital image is represented as:

$$f(x, y) = \begin{bmatrix} f(1,1) & f(1,2) & \dots & f(1,N) \\ f(2,1) & f(2,2) & \dots & f(2,N) \\ \vdots & \vdots & \vdots & \vdots \\ f(M,1) & f(M,2) & \dots & f(M,N) \end{bmatrix}$$

In this representation, you can notice the shift in origin.

Reading images. Images are read in MATLAB environment using the function 'imread.' Syntax of imread is:

```
imread('filename');
```

where 'filename' is a string having the complete name of the image, including its extension.

For example,

```
>>F = imread(Penguins_grey.jpg);
>>G = imread(Penguins_RGB.jpg);
```

Please note that when no path information is included in 'filename,' 'imread' reads the file from the current directory. When an image from another directory has to be read, the path of the image has to be specified.

Fig. 1: Grayscale image of penguins

Fig. 2: RGB image of penguins

Semicolon (;) at the end of a statement is used to suppress the output. If it is not included, MATLAB displays on the screen the result of the operation specified in that line.

'>>' indicates the beginning of a command line as it appears in the MATLAB command window.

Figs 1 and 2 show grayscale and RGB images of penguins, respectively. These images can be downloaded from the EFY website and stored in the current working directory.

imread, imshow and imwrite functions in MATLAB are used to read images in MATLAB environment, display them on MATLAB desktop and write them to the current directory, respectively

In case of grayscale images, the resultant matrix of the imread statement comprises 256x256 or 65,536 elements. The first statement takes grey values of all the pixels in the grayscale image and puts them into a matrix F(256x256 elements), which is now a MATLAB variable on which various matrix operations can be performed.

In case of RGB images, pixel values now consist of a list of three values, giving red, green and blue components of the colour of the given pixel. Matrix 'G' is a three-dimensional matrix 256x256x3. If there is no semicolon, the result of the command is displayed on the screen.

In nutshell, the imread function reads pixel values from an image file and returns a matrix of all pixel values.

To get the size of a 2D image, you can write the command:

```
[M,N] = size(f)
```

This syntax returns the number of rows (M) and columns (N) in the image.

You can find additional information about the array using 'whos' command.

'whos f' gives name, size, bytes, class and attributes of the array 'f.'

As an example, 'whos F' in our case gives:

```
Name Size        Bytes    Class  Attributes
F    768×1024×3  2359296  uint8
```

'whos G' gives:

```
Name Size        Bytes    Class  Attributes
G    768×1024    786432   uint8
```

Image display. Images are displayed on the MATLAB desktop using the function imshow, which has the basic syntax:

```
imshow(f)
```

where 'f' is an image array of data type 'uint8' or double. Data type 'uint8' restricts the values of integers between 0 and 255.

It should be remembered that for a matrix of type 'double,' the imshow function expects the values to be between 0 and 1, where 0 is displayed as black and 1 as white. Any value between 0 and 1 is displayed as grayscale. Any value greater than 1 is displayed as white, and a value